Dr. Michael O. Oni

Applied Mathematician · Heat Transfer & Scientific Machine Learning

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Google Scholar · Scopus
Web of Science X-7347-2019
ResearchGate

859 citations **18** h-index **31** i10-index **68** publications

Google Scholar, October 2026. Web of Science: 454 citations, h-index 14.

Research interests

Computational mathematics, fluid dynamics and heat transfer; natural and mixed convection in channels and annuli; electroosmotic and electromagnetohydrodynamic microflows; entropy generation; non-Fourier (dual- and three-phase-lag) heat conduction; spectral and semi-analytical methods; physics-informed neural networks and scientific machine learning.

Appointments

2026 – present	**Postdoctoral Fellow**, Unit for Data Science and Computing, North-West University, South Africa
2025 – present	**Senior Lecturer**, Department of Mathematics, Ahmadu Bello University, Zaria
2022 – 2025	**Lecturer I**, Ahmadu Bello University, Zaria
2019 – 2022	**Lecturer II**, Ahmadu Bello University, Zaria
2016 – 2019	**Assistant Lecturer**, Ahmadu Bello University, Zaria
2014 – 2016	**Graduate Assistant**, Ahmadu Bello University, Zaria

Education

2021	**PhD Mathematics**, Ahmadu Bello University, Zaria. *Thesis: A Study on Electroosmotic Natural and Mixed Convection Flows in Vertical Microgeometries with Isothermal and Isoflux Boundary Conditions*
2016	**MSc Mathematics**, Ahmadu Bello University, Zaria. *Dissertation: A Study on Natural Convection Flow in Cylindrical Geometries with Time-Periodic Heat Input*
2012	**BSc (Hons) Mathematics, First Class**, Ahmadu Bello University, Zaria

Grants, awards & innovation

2025	Principal Investigator, TETFund Institution-Based Research (IBR) Grant, Ahmadu Bello University
2015	Innovation: the *paraletrix*, a generalization of the rhotrix (*Afrika Matematika* 26, 871–885)
2012	Vice-Chancellor's Award of Excellence, ABU 2011/2012 Convocation
2012	Bronze medal, National Mathematics Competition for University Students (NAMCUS), National Mathematical Centre, Abuja

Research supervision

PhD	Usman Rilwan · Electroosmotic natural convection of nanofluids with induced magnetic field · *Major supervisor, ongoing*
PhD	Luqman Azeez · Natural convection of nanofluids in a vertical channel with suction/injection · *Co-supervisor, ongoing*
PhD	Ismaila Oyelade · Dual-phase-lag natural convection in a vertical channel · *Co-supervisor, ongoing*
MSc	Fatai Mohammed · Entropy generation with isothermal/isoflux conditions · *Major supervisor, ongoing*
MSc	Henry Ibrahim · Temperature-dependent viscosity in natural and mixed convection · *Major supervisor, ongoing*
MSc	Shakira Adebayo · Heat and mass transfer in unsteady EMHD natural convection · *Major supervisor, 2026*
MSc	Ahmed Abdulmalik · Hall current effects on free convection in a porous channel · *Major supervisor, 2026*
MSc	Vampah Tizhe Peter · Entropy generation in MHD porous-channel flow with Navier slip · *Co-supervisor, 2026*
MSc	Usman Rilwan · Time-dependent natural convection with heat source/sink · *Major supervisor, 2021*

Peer review

92 verified reviews of 65 manuscripts (Web of Science), including *Heat Transfer – Asian Research*, *ZAMM*, *Propulsion and Power Research*, *International Journal of Ambient Energy*, *Numerical Heat Transfer A & B*, *Journal of Thermophysics and Heat Transfer*, *International Journal of Numerical Methods for Heat & Fluid Flow*, *IEEE Transactions on Industrial Informatics*, *Physica Scripta* and *Engineering Applications of Computational Fluid Mechanics*.

Teaching

Undergraduate: Numerical Analysis I & II, Engineering Calculus I & II, Methods of Linear Algebra, Mathematical Methods, Vectors, Algebra, Trigonometry, Sets and Number Systems, Discrete Structures. Postgraduate: Research Methodology and ICT (MATH 812); Engineering Mathematics and Numerical Methods for the PGD Civil Engineering programme. Online tutor, ABU Distance Learning Centre (2020 – present).

Conference presentations

2024	Two papers on EMHD and electrokinetic convection. 10th International Aegean Conferences on Innovation Technologies & Engineering, İzmir, Türkiye
2024	Mixed convective rarefied gas flow with flow reversal and cubic density–temperature variation. 8th International Conference on Global Practice of Multidisciplinary Scientific Studies, Dubai, UAE
2022	Radially varying heat source/sink on MHD free convection in a concentric annulus. Ankara International Congress on Scientific Research VI, Türkiye (oral)
2021	Two oral presentations. 39th Annual Conference, Nigerian Mathematical Society, Redeemer's University, Ede
2016	Two papers. 35th Annual Conference, Nigerian Mathematical Society, FUT Minna
2015	52nd Annual National Conference, Mathematical Association of Nigeria, UNEC, Enugu

Memberships & service

2025 –	Nigerian Young Academy

2014 –	Mathematical Association of Nigeria
2015 – 2021	Nigerian Mathematical Society
2026	Developed an app for the Nigerian Young Academy celebrating members promoted to Professor

Publications (68)

68. S. Isa, **M. O. Oni**, A. Musa (2026). A Semi-Analytical Framework for Transient Electro-MHD Oil Transport in Subsurface Capillaries. *International Journal of Applied Mechanics and Engineering*. doi:10.59441/ijame/221546
67. U. S. Rilwan, **M. O. Oni**, B. K. Jha (2026). Analytical investigation of dual streaming potentials in electroosmotic magnetohydrodynamic nanofluid flow with induced magnetic field. *Next Materials*. doi:10.1016/j.nxmate.2026.103487
66. **M. O. Oni**, B. K. Jha, T. S. Yusuf, M. S. Shagari (2026). Analytical modeling of coupled electro-hydrodynamic and thermal transport in two-phase porous-fluid annular channels: enhanced oil recovery. *Computational Geosciences*. doi:10.1007/s10596-026-10484-9
65. G. Ojemeri, **M. O. Oni**, C. I. O. Gabriel, E. Omokhuale, H. Usman, T. A. Rufa'i (2026). Combined convection of an exothermic fluid with varying viscosity due to asymmetric/symmetric heating. *International Communications in Heat and Mass Transfer*. doi:10.1016/j.icheatmasstransfer.2026.111447
64. **M. O. Oni**, S. Y. Adebayo, T. S. Yusuf, B. K. Jha (2026). Computational Treatment of Unsteady Natural Convective and Mass Transfer Flow in a Vertical Microchannel Under Electroosmotic Effects. *ZAMM - Journal of Applied Mathematics and Mechanics / Zeitschrift für Angewandte Mathematik und Mechanik*. doi:10.1002/zamm.70452
63. U. S. Rilwan, **M. O. Oni**, B. K. Jha (2026). Dual streaming potential effects on EMHD mixed convective nanofluid flow with viscous dissipation and Joule heating. *Proceedings of the Institution of Mechanical Engineers, Part A: Journal of Power and Energy*. doi:10.1177/09576509261490836
62. S. Isa, **M. O. Oni**, A. Musa (2026). Entropy Generation and Mixed Convection in Vertical Annuli with Variable-Viscosity: Applications in EOR. *Thermal Advances*. doi:10.1016/j.thradv.2026.100174
61. S. S. Mohammed, A. A. Muhammad, M. Alansari, **M. O. Oni**, A. I. Tanko, A. Saliu (2026). Feng-Liu-type paired contractions in suprametric spaces with applications to fractional differential inclusion models for COVID-19. *Journal of Nonlinear Mathematical Physics*.
60. M. T. Akolade, M. F. Oyedotun, **M. O. Oni**, Y. O. Tijani, A. T. Adeosun, T. Sathiyaraj, ..., M. M. Awad (2026). Gradient-Based Optimization Modeling of Total Entropy and Heat Transfer in Exothermic Blood Analog Gold Nanoparticle Suspension. *Engineering Reports*. doi:10.1002/eng2.71011
59. **M. O. Oni**, A. A. Hussain, B. K. Jha (2026). Interplay of ion-slip and Hall currents on magnetohydrodynamic natural convection in a vertical microchannel with suction/injection. *Next Chemical Engineering, 3, 100067*.
58. U. S. Rilwan, **M. O. Oni**, H. M. Jibril, B. K. Jha (2026). Optimizing advanced microthermal systems: Synergistic effects of dual streaming potentials and induced magnetic fields in EMHD-driven hybrid flows. *Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering*. doi:10.1177/09544089261486669
57. A. A. Hussain, **M. O. Oni** (2026). Role of heat source/sink and chemical reaction on natural convection flow in a vertical porous microchannel with Hall current and ion slip. *International Journal of Mechanics and Materials in Design*. doi:10.1007/s10999-026-09946-w
56. M. T. Akolade, M. F. Oyedotun, **M. O. Oni**, A. T. Adeosun, M. B. Matadi (2026). Spectral and Taguchi approaches to entropy generation minimization of a particulate Casson-micropolar nanofluid suspension in an unsteady flow regime. *Results in Engineering, 112666*.
55. U. S. Rilwan, **M. O. Oni**, H. M. Jibril, B. K. Jha (2026). Synergistic effects of electro-magnetohydrodynamics and induced magnetic fields on the convective dynamics of Cu–SnO_2–SiO_2/water nanofluids. *Next Nanotechnology, 10, 100747*.

54. **M. O. Oni**, W. S. Zhao, D. Mothibi, H. Wang, S. H. Mthethwa, A. E. S. Ezugwu (2026). Thermodynamic assessment of 3D fractional dual-phase lag thermal transport in annular cooling architectures for 3D-stacked ICs. *International Journal of Heat and Mass Transfer, 268, 129066*.

53. **M. O. Oni**, S. Y. Adebayo, T. S. Yusuf, B. K. Jha (2025). Coupled electroosmotic-electromagnetohydrodynamic transport in a vertical microchannel with suction/injection: Heat and mass transfer analysis for Lab-on-a-Chip cooling applications. *Propulsion and Power Research*. doi:10.1016/j.jppr.2025.12.008

52. **M. O. Oni**, U. S. Rilwan (2025). Coupled thermal and electrokinetic effects on natural convection in electromagnetohydrodynamic flow. *Journal of Mathematics and Artificial Intelligence, 1(2), 91–115*.

51. U. S. Rilwan, **M. O. Oni**, H. M. Jibril, B. K. Jha (2025). Effects of joule heating and viscous dissipation on electromagneto-hydrodynamic flow in a microchannel with electroosmotic effect: Enhancement of MEMS cooling. *Proceedings of the Institution of Mechanical Engineers, Part N: Journal of Nanomaterials, Nanoengineering and Nanosystems*. doi:10.1177/23977914231217929

50. **M. O. Oni**, U. S. Rilwan (2025). Interplay of steady/unsteady Jeffrey EMHD nanofluid flow formation in a vertical channel with induced magnetic field. *Journal of the Korean Physical Society*. doi:10.1007/s40042-025-01472-w

49. S. F. H. Mohd Kanafiah, F. Fauzi, A. R. Mohd Kasim, S. Mohd Zokri, N. S. Nordin, M. Mokhtar, **M. O. Oni** (2025). Numerical analysis for the fluid flow and heat transfer of free convection flow in a porous medium under viscoelastic properties. *Semarak International Journal of Fundamental and Applied Mathematics, 5(1), 11–24*.

48. **M. O. Oni**, M. T. Akolade, G. Samaila, T. S. Yusuf, Y. O. Tijani, A. Yusuf, P. B. Malgwi (2025). Theory of conjugate mixed convection flow of hybridized ethylene glycol based nanoparticles with Joule heating. *Journal of Thermal Analysis and Calorimetry*. doi:10.1007/s10973-024-13875-1

47. U. S. Rilwan, **M. O. Oni**, B. K. Jha, H. M. Jibril (2024). Analysis of Joule heating and viscous dissipation on electromagnetohydrodynamic flow with electroosmotic effect in a porous microchannel: A heat transfer miniature enhancement. *Heat Transfer*. doi:10.1002/htj.22974

46. M. T. Akolade, G. Samaila, **M. O. Oni**, A. Yusuf, P. B. Malgwi, T. S. Yusuf, Y. O. Tijani (2024). Dynamics of accelerating and decelerating flows: A boundary layer examination with non-similar approach and convective boundary condition. *International Journal of Non-Linear Mechanics*. doi:10.1016/j.ijnonlinmec.2024.104675

45. **M. O. Oni**, B. K. Jha (2024). Impact of asymmetric zeta-potential on natural convection flow in a vertical microannulus with electroosmotic and Joule heating effects. *Proceedings of the Institution of Mechanical Engineers, Part N: Journal of Nanomaterials, Nanoengineering and Nanosystems*. doi:10.1177/23977914221103995

44. **M. O. Oni**, B. K. Jha, J. M. Abba, O. H. Adebayo (2024). Influence of Radially Varying Magnetic Fields and Heat Sources/Sinks on MHD Free-Convection Flow Within a Vertical Concentric Annulus. *Power Engineering and Engineering Thermophysics*. doi:10.56578/peet030103

43. **M. O. Oni**, B. K. Jha, J. M. Abba, B. I. Mundi, O. H. Adebayo (2024). Start-off MHD electrokinetic Couette flow in an annulus: A Riemann-sum approximation approach. *International Journal of Applied Mechanics and Engineering*. doi:10.59441/ijame/190397

42. A. M. Obalalu, **M. O. Oni**, U. Khan, A. Abbas, T. Muhammad, A. Zaib (2024). Two-Phase Numerical Simulation for the Heat and Mass Transfer Evaluation Across a Vertical Deformable Sheet with Significant Impact of Solar Radiation and Heat Source/Sink. *Arabian Journal for Science and Engineering*. doi:10.1007/s13369-023-08585-z

41. B. K. Jha, **M. O. Oni** (2024). Unsteady natural convection flow in a vertical tube inspired by ramped surface temperature of the tube. *International Journal of Magnetics and Electromagnetism, 10(1)*.

40. **M. O. Oni**, B. K. Jha (2023). Entropy generation analysis of electroosmotic mixed convection flow in vertical microannulus with asymmetric heat fluxes. *International Communications in Heat and Mass Transfer*. doi:10.1016/j.icheatmasstransfer.2023.106813

39. M. S. Shagari, A. T. Imam, U. A. Danbaba, J. Yahaya, **M. O. Oni**, A. A. Tijjani (2023). Existence of fixed points via C*-algebra-valued simulation functions with applications. *The Journal of Analysis*. doi:10.1007/s41478-022-00509-8

38. **M. O. Oni**, B. K. Jha (2023). Heat generation/absorption effect on mixed convection flow in a vertical channel filled with a nanofluid: Exact solution. *Journal of Oil and Gas Research Reviews, 3(1), 1–14*.

37. **M. O. Oni**, U. S. Rilwan (2023). Role of suction/injection on electromagnetohydrodynamics natural convection flow in a porous microchannel with electroosmotic effect. *International Journal of Applied Mechanics and Engineering*. doi:10.59441/ijame/173021

36. Y. O. Tijani, S. D. Oloniiju, **M. O. Oni**, M. T. Akolade (2023). Surface dynamics and thermophysical mechanics on Reiner-Philippoff fluid flow. *International Journal of Modern Physics B*. doi:10.1142/s0217979224503399

35. **M. O. Oni**, B. K. Jha (2022). Analysis of Transient Buoyancy/Electroosmotic Driven Flow in a Vertical Microannulus with Velocity-Slip and Temperature-Jump. *Engineering Science & Technology*. doi:10.37256/est.3120221293

34. **M. O. Oni**, B. K. Jha, A. O. Ajibade (2022). Interplay of dual streaming potentials on electroosmotic mixed convection flow in a vertical microannulus with Joule heating effect. *International Communications in Heat and Mass Transfer*. doi:10.1016/j.icheatmasstransfer.2021.105839

33. M. O. Lawal, K. B. Kasali, H. A. Ogunseye, **M. O. Oni**, Y. O. Tijani, Y. T. Lawal (2022). On the mathematical model of Eyring–Powell nanofluid flow with non-linear radiation, variable thermal conductivity and viscosity. *Partial Differential Equations in Applied Mathematics*. doi:10.1016/j.padiff.2022.100318

32. **M. O. Oni**, A. B. Yusuf, T. S. Yusuf, O. H. Adebayo, L. A. Azeez (2022). Unsteady Generalized Couette Flow in a Horizontal Channel with Sudden Application or Removal of a Porous Material. *Contemporary Mathematics*. doi:10.37256/cm.3220221365

31. B. K. Jha, A. J. Chitumu, **M. O. Oni** (2021). Analysis of Transient Natural Convective Flow of a Nanofluid in a Vertical Tube. *International Journal of Applied Mechanics and Engineering*. doi:10.2478/ijame-2021-0018

30. B. K. Jha, **M. O. Oni** (2021). Joule heating and viscous dissipation effect on electroosmotic mixed convection flow in a vertical microchannel subjected to asymmetric heat fluxes. *Propulsion and Power Research*. doi:10.1016/j.jppr.2021.01.001

29. **M. O. Oni**, B. K. Jha (2020). Electroosmotic natural convection flow in a vertical microchannel with asymmetric heat fluxes. *SN Applied Sciences*. doi:10.1007/s42452-020-03413-7

28. **M. O. Oni**, B. K. Jha (2020). Heat transfer analysis of mixed convection flow in a vertical microchannel with electrokinetic effect. *GEM - International Journal on Geomathematics*. doi:10.1007/s13137-020-00165-9

27. (2020). Impact of Space-Dependent Thermal Conductivity on Heat Transfer in a Vertical Annulus with Asymmetric Surface Heating. *Significances of Bioengineering & Biosciences*. doi:10.31031/sbb.2020.04.000578

26. B. K. Jha, **M. O. Oni** (2020). Nonlinear Mixed Convection Flow in an Inclined Channel with Time-Periodic Boundary Conditions. *International Journal of Applied and Computational Mathematics*. doi:10.1007/s40819-020-00880-9

25. **M. O. Oni**, B. K. Jha (2019). Heat generation/absorption effect on natural convection flow in a vertical annulus with time-periodic boundary conditions. *Journal of Aircraft and Spacecraft Technology, 3(1), 183–196*.

24. B. K. Jha, **M. O. Oni** (2019). Mathematical modelling of combined pressure driven and electrokinetic effect in a channel with induced magnetic field: An exact solution. *Journal of King Saud University - Science*. doi:10.1016/j.jksus.2018.10.009

23. B. K. Jha, **M. O. Oni** (2019). Natural convection flow in a vertical annulus with time-periodic thermal boundary conditions. *Propulsion and Power Research*. doi:10.1016/j.jppr.2018.12.002

22. B. K. Jha, **M. O. Oni** (2019). Role of Sudden Application or Withdrawal of Magnetic Field on MHD Couette Flow. *International Journal of Applied Mechanics and Engineering*. doi:10.2478/ijame-2019-0051

21. B. K. Jha, **M. O. Oni** (2019). Steady Fully Developed Mixed Convection Flow in a Vertical Channel with Heat and Mass Transfer and Temperature-Dependent Viscosity: An Exact Solution. *Zeitschrift für Naturforschung A: A Journal of Physical Sciences*. doi:10.1515/zna-2018-0367

20. **M. O. Oni**, B. K. Jha (2019). Theoretical analysis of transient natural convection flow in a vertical microchannel with electrokinetic effect. *Journal of Taibah University for Science*. doi:10.1080/16583655.2019.1682830
19. B. K. Jha, **M. O. Oni** (2019). Theory of fully developed mixed convection including flow reversal: A nonlinear Boussinesq approximation approach. *Heat Transfer - Asian Research*. doi:10.1002/htj.21550
18. B. K. Jha, L. A. Azeez, **M. O. Oni** (2019). Unsteady hydromagnetic-free convection flow with suction/injection. *Journal of Taibah University for Science*. doi:10.1080/16583655.2018.1545624
17. B. K. Jha, **M. O. Oni** (2018). An analytical solution for temperature field around a cylindrical surface subjected to a time dependent heat flux: An alternative approach. *AEJ - Alexandria Engineering Journal*. doi:10.1016/j.aej.2017.01.003
16. B. K. Jha, **M. O. Oni** (2018). Electromagnetic natural convection flow in a vertical microchannel with Joule heating: exact solution. *Journal of Taibah University for Science*. doi:10.1080/16583655.2018.1494423
15. T. Yusuf, **M. O. Oni** (2018). Entropy generation under the influence of radial magnetic field and viscous dissipation of generalized Couette flow in an annulus. *Propulsion and Power Research*. doi:10.1016/j.jppr.2018.11.005
14. B. K. Jha, **M. O. Oni** (2018). Fully developed mixed convection flow in a vertical channel with electrokinetic effects: Exact solution. *Multidiscipline Modeling in Materials and Structures*. doi:10.1108/mmms-10-2017-0123
13. B. K. Jha, **M. O. Oni** (2018). Impact of mode of application of magnetic field on rate of heat transfer of rarefied gas flows in a microtube. *AEJ - Alexandria Engineering Journal*. doi:10.1016/j.aej.2017.03.029
12. B. K. Jha, **M. O. Oni** (2018). Mixed convection flow in a vertical channel with temperature dependent viscosity and flow reversal: An exact solution. *International Journal of Heat and Technology*. doi:10.18280/ijht.360225
11. B. K. Jha, **M. O. Oni** (2018). Natural convection flow in a vertical micro-annulus with time-periodic thermal boundary conditions: An exact solution. *Multidiscipline Modeling in Materials and Structures*. doi:10.1108/mmms-09-2017-0111
10. B. K. Jha, **M. O. Oni** (2018). Role of magnetic field on mixed convective flow between two concentric microtubes with heat generation/absorption: An exact solution. *Proceedings of the Institution of Mechanical Engineers, Part N: Journal of Nanomaterials, Nanoengineering and Nanosystems*. doi:10.1177/2397791418776520
9. B. K. Jha, **M. O. Oni**, B. Aina (2018). Steady fully developed mixed convection flow in a vertical micro-concentric-annulus with heat generating/absorbing fluid: An exact solution. *Ain Shams Engineering Journal*. doi:10.1016/j.asej.2016.08.005
8. B. K. Jha, **M. O. Oni** (2018). Transient natural convection flow between vertical concentric cylinders heated/cooled asymmetrically. *Proceedings of the Institution of Mechanical Engineers, Part A: Journal of Power and Energy*. doi:10.1177/0957650918758743
7. **M. O. Oni** (2017). Combined effect of heat source, porosity and thermal radiation on mixed convection flow in a vertical annulus: An exact solution. *Engineering Science and Technology, an International Journal*. doi:10.1016/j.jestch.2016.12.009
6. B. K. Jha, **M. O. Oni** (2017). Laminar Convection in a Uniformly Heated Vertical Porous Channel Revisited. *Advances in Applied Sciences*. doi:10.11648/j.aas.20170203.11
5. **M. O. Oni**, T. S. Yusuf (2017). Unsteady Couette flow in an annulus with combined mode of magnetic field application: A generalization. *Mathematical Modelling of Engineering Problems, 4(4), 168–172*.
4. B. K. Jha, **M. O. Oni** (2016). Natural convection flow in a vertical tube inspired by time-periodic heating. *AEJ - Alexandria Engineering Journal*. doi:10.1016/j.aej.2016.08.025
3. A. Aminu, **M. O. Oni** (2015). An introduction to the concept of paraletrix, a generalization of rhotrix. *Afrika Matematika, 26(5–6), 871–885*. doi:10.1007/s13370-014-0251-1
2. **M. O. Oni**, A. Aminu (2015). The exponential paraletrix. *Journal of the Nigerian Association of Mathematical Physics, 31, 271–278*.
1. A. Aminu, **M. O. Oni** (2013). Adjacent rhotrix of a complete, simple and undirected graph. *Journal of the Nigerian Association of Mathematical Physics, 25(2), 267–274*.

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